



# **EWM22A-400/900BWL22S Series User Manual**

**2.4GHz 400/900MHz 20/22dBm BLE/WIFI/LoRa All-in-One Wireless  
Module**



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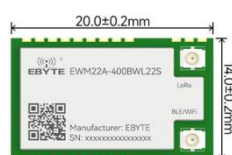
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# 1 Introduction

## 1.1 Brief Introduction

EWM22A-400/900BWL22S is a BLE, WIFI, LoRa all-in-one wireless data transmission module developed by Chengdu Yibert, this series of modules support BLE, WIFI, LoRa communication, of which the LoRa function is compatible with the E22-T series of modules, SMD-type, the maximum transmit power of 20dBm for BLE/WIFI and 22dBm for LoRa. Transmit power, pin spacing 1.27mm, factory self-firmware, powerful, mainly for data transmission application scenarios.



EWM22A-400BWL22S



EWM22A-900BWL22S

## 1.2 Features

### Product Functional Features:

- Supports LoRa, WIFI (IEEE 802.11b/g/n), and BLE5.0 all-in-one communication technology, which makes the applicable scenes more diversified;
- Supports BLE over-the-air firmware encryption upgrade, which is convenient for later maintenance;
- Supports AT command, which is more convenient to use;
- Supports deep hibernation, and the whole power consumption in this mode is about 6.7uA;
- Parameters are saved when power down, and the module will work according to the set parameters after power up again;
- High-efficiency watchdog design, once an abnormality occurs, the module will restart automatically, and can continue to work according to the previous parameter settings;
- 4 MB embedded FLASH;
- Flexible deployment: a single module supports multiple protocols, reducing hardware redundancy and lowering IoT system integration costs.

### LoRa Functional Features:

- Supports AT command and HEX command settings;
- Adopts the new generation LoRa spread spectrum modulation technology, which brings longer communication distance and stronger anti-interference ability;
- Supports automatic relay networking, and multi-level relay is suitable for ultra-long-distance communication, and the same area runs multiple networks at the same time;
- Supports users to set their own communication key and cannot be read, which greatly improves the confidentiality of user data;
- Supports LBT function to listen to the channel environmental noise before sending, which can greatly improve the

success rate of communication of the module in harsh environments;

- Supports RSSI signal strength indication function, which is used to evaluate signal quality, improve communication network, and range measurement;
- Supports wireless parameter configuration, sending command packets wirelessly to remotely configure or read wireless module parameters;
- Supports Wake-on-Air, i.e., ultra-low power consumption function for battery-powered application solutions;
- Supports fixed-point transmission, broadcast transmission, and channel listening;
- Communication distance up to 5km under ideal conditions;
- Supports data transmission rates from 2.4K to 62.5Kbps;
- Industrial-grade standard design, support -40 ~ +85 °C under long time use;
- Module power can be up to 160mW (22dBm), the transmission is more far and stable.

BLE Functional Features:

- Supports the slave role;
- Supports the Bluetooth BLE 5.0 protocol;
- Supports Bluetooth parameter over-the-air configuration function;
- Supports transmit power modification with a maximum transmit power of 20dBm;
- Maximum communication maximum distance up to 500m;
- MTU max 517 bytes;

WIFI functionality features:

- Supports STA working mode;
- Supports disconnected automatic reconnection;
- Supports WPA2 WIFI security authentication method;
- Supports TCP CLIENT communication mode;
- Transmit power is 20dBm;
- The maximum communication distance of is up to 500m;

## 1.3 Application

- Smart home as well as industrial sensors, etc;
- Security systems, positioning systems;
- Wireless remote control, drones;
- Healthcare products;
- Automotive industry applications

## 2 Specification

### 2.1 Limit parameters

|  |                     |  |
|--|---------------------|--|
|  | <b>Performances</b> |  |
|--|---------------------|--|

| Main Parameters            | Mini | Max | Remark                                        |
|----------------------------|------|-----|-----------------------------------------------|
| Power supply voltage (V)   | 0    | 3.6 | Permanent module burnout above 3.6V           |
| Blocking power (dBm)       | -    | 10  | Low probability of burnout in close proximity |
| Operating temperature (°C) | -40  | +85 | Industrial Grade                              |

## 2.2 Operating parameters

| Main Parameters                                            |                         | Performances |         |         | Remark                                                     |
|------------------------------------------------------------|-------------------------|--------------|---------|---------|------------------------------------------------------------|
|                                                            |                         | Mini         | typical | Max     |                                                            |
| Operating Voltage (V)                                      |                         | 3.0          | 3.3     | 3.6     | ≥3.3V can guarantee output power                           |
| Communication level (V)                                    |                         |              | 3.3     |         | Risk of burn-in with 5V TTL                                |
| Operating temperature (°C)                                 |                         | -40          | -       | +85     | Industrial grade design                                    |
| BLE/WIFI operating band (GHz)                              |                         | 2.400        | -       | 2.480   | Supports ISM bands                                         |
| EWM22A-400BWL22S<br>LoRa<br>Operating Frequency Band (MHz) |                         | 410.125      | 433.125 | 493.125 |                                                            |
| EWM22A-900BWL22S<br>LoRa<br>Operating frequency band (MHz) |                         | 850.125      | 868.125 | 930.125 |                                                            |
| LoRa air rate (Kbps)                                       |                         | 2.4          | 62.5    | 62.5    |                                                            |
| power consumption                                          | Transmit Current (mA)   |              | 185     |         | Mode 1, BLE listening off, instantaneous power consumption |
|                                                            | Receive Current (mA)    |              | 23      |         | Mode 1, BLE listening off, average power consumption       |
|                                                            | Deep sleep current (μA) |              | 6.7     |         |                                                            |
| BLE transmit power (dBm)                                   |                         | -9           | 20      | 20      | Configurable                                               |
| WIFI transmit power (dBm)                                  |                         | -            | 20      | -       | Fixed, not configurable                                    |
| LoRa transmit power (dBm)                                  |                         | 10           | 22      | 22      | Configurable                                               |

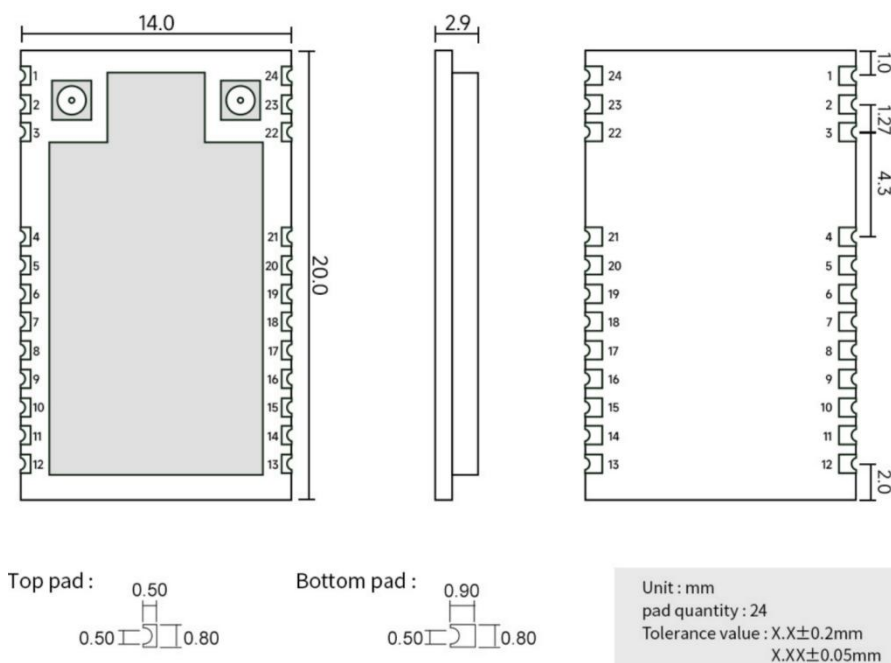
| Main Parameters             | Description | Remark                                                            |
|-----------------------------|-------------|-------------------------------------------------------------------|
| BLE communication distance  | 500m        | Clear and open environment, antenna gain 5.0dBi, height 2 meters. |
| WIFI communication distance | 500m        | Clear and open environment, antenna gain 5.0dBi, height 2 meters. |

|                             |                 |                                                                                     |
|-----------------------------|-----------------|-------------------------------------------------------------------------------------|
| LoRa communication distance | 5Km             | Clear and open environment, antenna gain 3.5dBi, height 2 meters, air rate 2.4kbps. |
| Protocols                   | BLE/WIFI/LoRa   | BLE5.0、Wi-Fi(IEEE 802.11b/g/n)                                                      |
| Communication Interface     | UART            | TTL level                                                                           |
| Packaging                   | SMD             |                                                                                     |
| LoRa packet length          | 240 Bbytes      | Command-set packetized 32/64/128/240 byte sends                                     |
| BLE MTU                     | 517 Bbytes      |                                                                                     |
| UART RX BUFFER              | 1024 Bbytes     |                                                                                     |
| UART TX BUFFER              | 1024 Bbytes     |                                                                                     |
| LORA_TX_BUF_SIZE            | 2028 Bbytes     |                                                                                     |
| BLE_TX_BUF_SIZE             | 1024 Bbytes     |                                                                                     |
| TCP_TX_BUF_SIZE             | 2048 Bbytes     |                                                                                     |
| Interface                   | 1.27mm          | Stamp Hole                                                                          |
| Dimension                   | 14*20mm         | ±0.2mm                                                                              |
| LoRa Antenna Interface      | IPEX/Stamp Hole | Equivalent impedance about 50Ω                                                      |
| BLE/WIFI Antenna Interface  | IPEX            |                                                                                     |
| Weight                      | 1.2g            | ±0.1g                                                                               |



## 3 Mechanical Dimensions and Pin Definitions

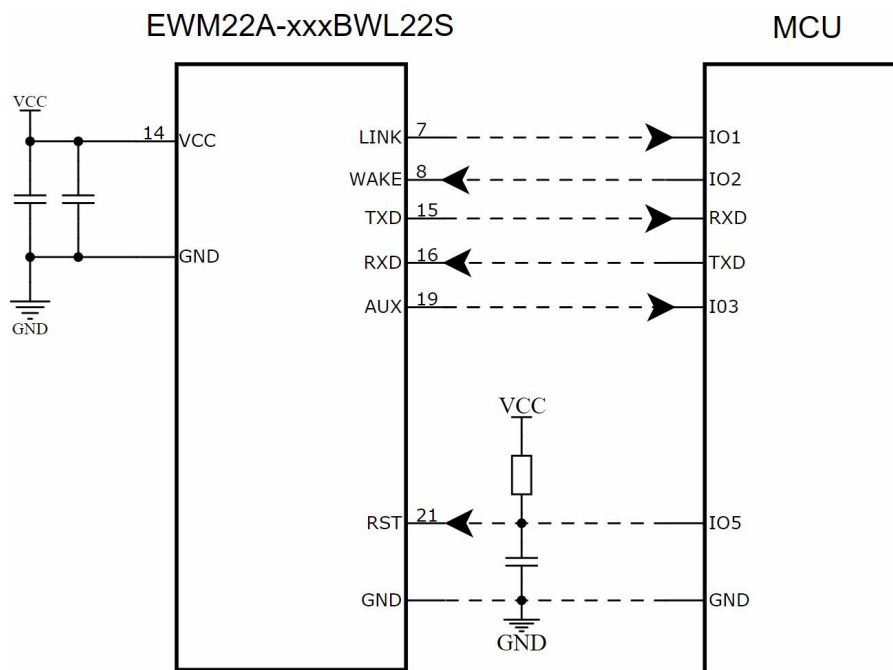
### 3.1 Dimensions and Pin Definitions



| Pin No. | Pin Name | Pin Direction | Pin Purpose                                                                                                                                                                                                                 |
|---------|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | GND      | -             | Power Ground                                                                                                                                                                                                                |
| 2       | ANT      | -             | LoRa antenna interface, 50 ohm characteristic impedance                                                                                                                                                                     |
| 3       | GND      | -             | Power Ground                                                                                                                                                                                                                |
| 4       | GND      | -             | Power Ground                                                                                                                                                                                                                |
| 5       | TX_EN    | -             | Reserved Pins                                                                                                                                                                                                               |
| 6       | RX_EN    | -             | Reserved Pins                                                                                                                                                                                                               |
| 7       | LINK     | Output        | Bluetooth/WIFI connection status indication pin:<br>BLE:Default high,BLE connected is low;<br>WIFI: default high, WIFI connected, TCP not connected outputs 1Hz high and low level change, TCP connected outputs low level. |
| 8       | WAKE     | Input         | Deep Hibernation Wake-Up Pin, Wake-Up Low.                                                                                                                                                                                  |
| 9       | NC       | Input/Output  | Reserved Pins                                                                                                                                                                                                               |
| 10      | NC       | Input/Output  | Reserved Pins                                                                                                                                                                                                               |
| 11      | NC       | Input/Output  | Reserved Pins                                                                                                                                                                                                               |
| 12      | NC       | Input/Output  | Reserved Pins                                                                                                                                                                                                               |
| 13      | GND      | -             | Power Ground                                                                                                                                                                                                                |
| 14      | VCC      | Input         | Power input: 3.0~3.6V                                                                                                                                                                                                       |
| 15      | TXD      | Output        | TTL serial output connected to external RXD input pin;                                                                                                                                                                      |
| 16      | RXD      | Input         | TTL serial input to external TXD output pin;                                                                                                                                                                                |
| 17      | NC       | -             | Reserved pins                                                                                                                                                                                                               |

|    |     |        |                                                                                                                                                     |
|----|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 18 | NC  | -      | Reserved pins                                                                                                                                       |
| 19 | AUX | Output | Used to indicate LoRa operating status; user wakes up external MCU and outputs low during power-on self-test initialization; (can be left dangling) |
| 20 | NC  | -      | Reserved pins                                                                                                                                       |
| 21 | RST | Input  | Reset pin, low level reset                                                                                                                          |
| 22 | GND | -      | Power ground                                                                                                                                        |
| 23 | NC  | -      | Reserved pin                                                                                                                                        |
| 24 | GND | -      | Power Ground                                                                                                                                        |

## 3.2 Recommended Wiring Diagram



## 4 Function Introduction

### 4.1 Working mode

The module supports four communication modes: BLE, WIFI, LoRa and URAT, and implements 8 modes together. Each mode includes UART function (for mode switching, data sending and receiving and monitoring), and BLE and WIFI cannot coexist. The 8 working modes are as follows:

| Working mode                                 | Description                                                                                                                                                                                                                                                          |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 --- Configuration Mode                     | Enable UART, BLE function<br>1、You can use AT instruction to configure the module through BLE and UART.                                                                                                                                                              |
| 1 --- UART/LORA pass-through, BLE listening  | Enable UART, BLE, LORA function<br>1、BLE listens to the data flow direction and outputs the corresponding data;<br>2、Can use AT commands to configure the module via BLE;<br>3、Can receive remote LoRa configuration commands;                                       |
| 2 --- UART/BLE passthrough, LORA listening   | Enable UART, BLE, LORA function<br>1.LoRa listens to the data flow direction and outputs the corresponding data;<br>2. can use AT commands to configure the module via BLE;<br>3. Can receive remote LoRa configuration commands;                                    |
| 3 --- LORA/BLE passthrough, UART listening   | Enable UART, BLE, LORA function<br>1、UART listens to the data flow direction and outputs the corresponding data;<br>2、Can use AT command to configure the module via BLE;<br>3、Can receive remote LoRa configuration commands;                                       |
| 4 --- UART/LORA pass-through, WIFI listening | Enable UART, WIFI, LORA function<br>1、After WIFI connection, the tcp server listens to the data flow direction and outputs the corresponding data;<br>2、You can use AT command to configure the module via TCP;<br>3、Can receive remote LoRa configuration commands; |
| 5 --- LORA/WIFI pass-through, UART listening | Enable UART, WIFI, LoRa function<br>1、UART listens to the data flow direction and outputs the corresponding data;<br>2、Can use AT command to configure the module via TCP;                                                                                           |

|                                              |                                                                                                                                                                                                                                        |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | 3、 Can receive remote LoRa configuration commands;                                                                                                                                                                                     |
| 6 --- UART/WIFI pass-through, LORA listening | Enable UART, WIFI, LoRa function<br>1、 LoRa listens to the data flow direction and outputs the corresponding data;<br>2、 Can use AT commands to configure the module via TCP;<br>3、 Can receive remote LoRa configuration commands;    |
| 7 --- LORA/UART pass-through, WOR mode       | Open UART, LoRa function<br>1, WOR slave timed wake-up scanning to determine whether there is a host to send data, there is data to receive data and wake up the module and receive data, there is no data to enter hibernation again; |

## 4.2 BLE Functions in Detail

| UUID | Description            | Remark                    |
|------|------------------------|---------------------------|
| FFF1 | Transmission Receiving | NOTIFY                    |
| FFF2 | Transmit-Transmit      | WRITE, WRITE_NO_RESPONSE  |
| FFF3 | AT Configuration       | NOTIFY, WRITE_NO_RESPONSE |

Basic functions:

- 1.BLE only supports slave, you can configure the module and send/receive serial data through APP;
2. module firmware upgrade through BLE OTA.APK;
3. Only support AT command to configure BLE related configuration;

Default parameters:

1. device name: EWM22A-400BWL22S/EWM22A-900BWL22S (module model);
  2. broadcast interval: 100ms;
  3. Connection interval 20ms~40ms.
  4. Broadcast type is connectable and scannable broadcast;
  5. Connection timeout 2.5 seconds;
- UUID is 16 bits by default;

## 4.3 WIFI/TCP Functions in Detail

Basic functions:

- 1.WIFI is only supported as station;
- 2.TCP only supports Ipv4;
3. Only AT commands are supported for configuring WIFI-related configurations;

## 4.4 LoRa Features in Detail

Basic function: Support AT instruction and HEX instruction setting

#### 4.4.1 LoRa HEX command format

The list of supported commands in the configuration mode (Mode 0) is as follows (serial port parameters are only supported in 9600, 8N1 format if setup with the E22 host computer):

| No. | command format             | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Setup Register             | <p>Instruction: C0 + start address + length + parameters<br/>Response: C1 + start address + length + parameters</p> <p>Example 1: Configure the channel as 0x09<br/>Command Start address Length Parameters<br/>Send: C0 05 01 09<br/>Return: C1 05 01 09</p> <p>Example 2: Configure module address (0x1234), network address (0x00), serial port (9600 8N1) and airspeed (2.4K) at the same time<br/>Send: C0 00 04 12 34 00 62<br/>Return: C1 00 04 12 34 00 62</p>                         |
| 2   | Read register              | <p>Command: C1 + start address + length<br/>Response: C1 + start address + length + parameters</p> <p>Example 1: Read channel<br/>Command Start address Length Parameters<br/>Send: C1 05 01<br/>Return: C1 05 01 09</p> <p>Example 2: Read module address, network address, serial port, airspeed at the same time<br/>Send: C1 00 04<br/>Return: C1 00 04 12 34 00 61</p>                                                                                                                    |
| 3   | Setting Temporary Register | <p>Command: C2 + start address + length + parameters<br/>Response: C1 + start address + length + parameters</p> <p>Example 1: Configure channel as 0x09<br/>Command Start address Length Parameters<br/>Send: C2 05 01 09<br/>Return: C1 05 01 09</p> <p>Example 2: Configure module address (0x1234), network address (0x00), serial port (9600 8N1), and airspeed (2.4K) simultaneously<br/>Send: C2 00 04 12 34 00 62<br/>Return: C1 00 04 12 34 00 62</p>                                  |
| 4   | Wireless Configuration     | <p>Command: CF CF + regular command<br/>Response: CF CF + regular response</p> <p>Example 1: Wireless configuration channel is 0x09<br/>Wireless Command Header Command Start Address Length Parameters<br/>Send: CF CF C0 05 01 09<br/>Return: CF CF C1 05 01 09</p> <p>Example 2: Wireless configure module address (0x1234), network address (0x00), serial port (9600 8N1), airspeed (2.4K) simultaneously<br/>Send: CF CF C0 00 04 12 34 00 62<br/>Return: CF CF C1 00 04 12 34 00 62</p> |
| 5   | Formatting error           | <p>Format Error Response<br/>FF FF FF FF</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

#### 4.4.2 LoRa register description

| No  | Read<br>Write  | Name                | Description      |                |                        |                                      | Remark                                                                                                                                                                                                                                                                                                                                                                          |   |
|-----|----------------|---------------------|------------------|----------------|------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 00H | Read<br>/Write | ADDH                | ADDH（default 0）  |                |                        |                                      | Module address high byte and low byte;<br>Note: When the module address is equal to FFFF, it can be used as the broadcast and listen address, i.e.: at this point the module will not perform address filtering                                                                                                                                                                 |   |
| 01H | Read<br>/Write | ADDL                | ADDL（default 0）  |                |                        |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
| 02H | Read<br>/Write | NETID               | NETID（default 0） |                |                        |                                      | Network address for distinguishing networks;<br>When communicating with each other, they should be set to the same.                                                                                                                                                                                                                                                             |   |
| 03H | Read<br>/Write | EG0                 | 7                | 6              | 5                      | UART serial port rate (bps)          | Two modules communicating with each other can have different serial port baud rates and different checksums;<br><br>When transmitting larger data packets continuously, the user needs to consider the data blocking brought by the same baud rate, and may even be lost;<br><br>It is generally recommended that the baud rates of the two communicating parties are the same. |   |
|     |                |                     | 0                | 0              | 0                      | Serial port baud rate of 1200        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 0              | 1                      | Serial port baud rate of 2400        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 1              | 0                      | Serial port baud rate of 4800        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 1              | 1                      | Serial port baud rate of 9600        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 0              | 0                      | Serial port baud rate of 19200       |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 0              | 1                      | Serial Baud Rate of 38400            |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 1              | 0                      | Serial baud rate is 57600            |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 1              | 1                      | Serial baud rate is 115200 (default) |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 4                | 3              | Serial port parity bit |                                      | The serial port modes can be different on both sides of the communication;                                                                                                                                                                                                                                                                                                      |   |
|     |                |                     | 0                | 0              | 8N1 (default)          |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 1              | 8O1                    |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 0              | 8E1                    |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 1              | 8N1 (00 equivalent)    |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 2                | 1              | 0                      | Wireless Air Rate (bps)              | The air rate must be the same on both sides of the communication;<br><br>The higher the air rate, the lower the delay and the shorter the transmission distance.                                                                                                                                                                                                                |   |
|     |                |                     | 0                | 0              | 0                      | Air Rate 2.4k                        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 0              | 1                      | Air Rate 2.4k                        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 1              | 0                      | Air Rate 2.4k (default)              |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 0                | 1              | 1                      | Air Rate 4.8k                        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 0              | 0                      | Air Rate 9.6k                        |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 0              | 1                      | Air Rate 19.2k                       |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 1              | 0                      | Air Rate 38.4k                       |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 1                | 1              | 1                      | Airspeed 62.5k                       |                                                                                                                                                                                                                                                                                                                                                                                 |   |
|     |                |                     | 04H              | Read<br>/Write | REG1                   | 7                                    |                                                                                                                                                                                                                                                                                                                                                                                 | 6 |
| 0   | 0              | 240 bytes (default) |                  |                |                        |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
| 0   | 1              | 128 bytes           |                  |                |                        |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |
| 1   | 0              | 64 bytes            |                  |                |                        |                                      |                                                                                                                                                                                                                                                                                                                                                                                 |   |

|  |  |  |   |                                                                  |                 |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|--|--|---|------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 1 | 1                                                                | 32 bytes        | If the data sent by the user is larger than the packet length, the serial port at the receiving end will output in packets.                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 5 | RSSI Ambient Noise Enable                                        |                 |                                                                                                                                                                                  | When enabled, instruction C0 C1 C2 C3 can be sent in UART-IoRa mode or WOR transmit mode Instruction Read Register;<br>Register 0x00 : Current ambient noise RSSI;<br>Register 0X01 : RSSI of the last received data.<br>(The current channel noise is: dBm = -(256 - RSSI));<br>Instruction format: C0 C1 C2 C3 + start address + read length;<br>return: C1 + address + read length + read the effective value; for example: send C0 C1 C2 C3 00 01<br>Return C1 00 01 RSSI (address can only start from 00)<br><br>Translated with www.DeepL.com/Translator (free version) |
|  |  |  | 0 | Disable (default)                                                |                 |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 1 | Enable                                                           |                 |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 4 | 3                                                                | reservation     | reservation                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 2 | (E22 is software switching, EWM22 removes this feature) Retained |                 |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 1 | 0                                                                | Transmit power  | Power and current are non-linearly related, and the power supply is most efficient at maximum power;<br><br>Current does not decrease in the same proportion as power decreases. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 0 | 0                                                                | 22dBm (default) |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 0 | 1                                                                | 17dBm           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 1 | 0                                                                | 13dBm           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  |  |  | 1 | 1                                                                | 10dBm           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|     |             |      |                                                                                                                                                                |  |  |                                                                              |
|-----|-------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------|
| 05H | Read /Write | REG2 | Channel Control (CH)<br>0-83 represent a total of 84 channels (for 400 band) respectively<br>0-80 represent a total of 81 channels respectively (for 900 band) |  |  | Actual frequency = 410.125 + CH * 1M<br>Actual frequency = 850.125 + CH * 1M |
|-----|-------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------|

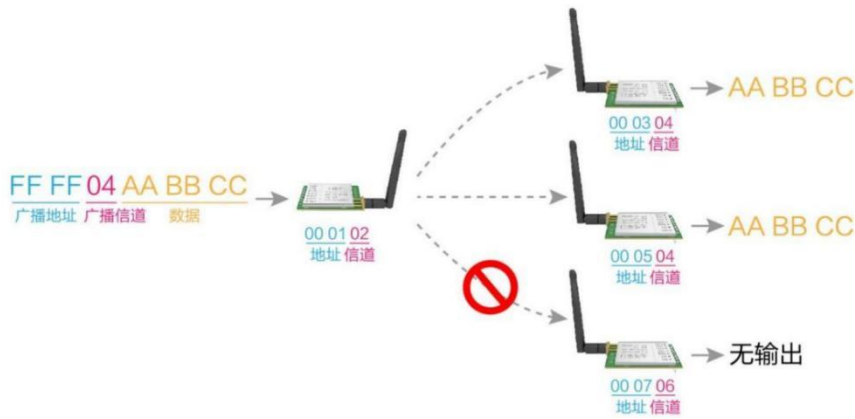
|     |                       |      |   |                                    |  |                                                                                                                                                                                                                                                                  |
|-----|-----------------------|------|---|------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06H | Read /Write           | REG3 | 7 | Enable RSSI bytes                  |  | When enabled, the module receives wireless data, which will follow an RSSI strength byte when output through the serial port TXD.                                                                                                                                |
|     |                       |      | 0 | Disable (default)                  |  |                                                                                                                                                                                                                                                                  |
|     |                       |      | 1 | Enable                             |  |                                                                                                                                                                                                                                                                  |
|     |                       |      | 6 | Transmission Method                |  | For fixed-point transmission, the module will recognize the first three bytes of the serial data as: address high + address low + channel, and will use it as the wireless transmit target.                                                                      |
|     |                       |      | 0 | Transparent transmission (default) |  |                                                                                                                                                                                                                                                                  |
|     |                       |      | 1 | Fixed transmission                 |  |                                                                                                                                                                                                                                                                  |
|     |                       |      | 5 | Relay Function                     |  | When the relay function is enabled, the module will initiate a forwarding if the destination address is not the module itself;<br>In order to prevent data backhauling, it is recommended to be used in conjunction with fixed-point mode; i.e., the destination |
|     |                       |      | 0 | Disable relay function (default)   |  |                                                                                                                                                                                                                                                                  |
| 1   | Enable relay function |      |   |                                    |  |                                                                                                                                                                                                                                                                  |

|         |       |         |                             |                                                                                                                                             |   |           |                                                                                                                                                                                                                                  |                                                                                                                                             |
|---------|-------|---------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|         |       |         |                             |                                                                                                                                             |   |           | address is different from the source address.                                                                                                                                                                                    |                                                                                                                                             |
|         |       |         | 4                           | LBT Enable                                                                                                                                  |   |           |                                                                                                                                                                                                                                  | When enabled, wireless data will be listened to before transmitting, which can avoid interference to some extent, but may bring data delay; |
|         |       |         | 0                           | Disable (default)                                                                                                                           |   |           |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 1                           | Enable                                                                                                                                      |   |           |                                                                                                                                                                                                                                  | The maximum dwell time of LBT is 2 seconds, and it will be forced to send out when it reaches two seconds.                                  |
|         |       |         | 3                           | WOR mode transceiver control                                                                                                                |   |           |                                                                                                                                                                                                                                  | Valid only for mode 7;                                                                                                                      |
|         |       |         | 0                           | WOR receiver (default)<br>Works in WOR listening mode, see below for listening period (WOR period), which saves a lot of power consumption. |   |           |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 1                           | WOR transmitter<br>The module transmits and receives open and adds a wake-up code for a certain period of time when transmitting data.      |   |           |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 2                           | 1                                                                                                                                           | 0 | WOR cycle |                                                                                                                                                                                                                                  | Valid only for mode 1;                                                                                                                      |
|         |       |         | 0                           | 0                                                                                                                                           | 0 | 500ms     |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 0                           | 0                                                                                                                                           | 1 | 1000ms    |                                                                                                                                                                                                                                  | Cycle time T= (1+WOR)*500ms, the maximum is 4000ms and the minimum is 500ms;                                                                |
|         |       |         | 0                           | 1                                                                                                                                           | 0 | 1500ms    |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 0                           | 1                                                                                                                                           | 1 | 2000ms    |                                                                                                                                                                                                                                  | The longer the period time between WOR listening intervals, the lower the average power consumption, but the higher the data delay;         |
|         |       |         | 1                           | 0                                                                                                                                           | 0 | 2500ms    |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 1                           | 0                                                                                                                                           | 1 | 3000ms    |                                                                                                                                                                                                                                  | Both sender and receiver must agree (very important)                                                                                        |
|         |       |         | 1                           | 1                                                                                                                                           | 0 | 3500ms    |                                                                                                                                                                                                                                  |                                                                                                                                             |
|         |       |         | 1                           | 1                                                                                                                                           | 1 | 4000ms    |                                                                                                                                                                                                                                  |                                                                                                                                             |
| 07H     | Write | CRYPT_H | Key high byte (default 0)   |                                                                                                                                             |   |           | Write only, read returns 0;                                                                                                                                                                                                      |                                                                                                                                             |
| 08H     | Write | CRYPT_L | Key low byte (default 0)    |                                                                                                                                             |   |           | Used for encryption to avoid interception of over-the-air wireless data by similar modules;<br>The module will internally use these two bytes as a calculation factor to transform and encrypt the over-the-air wireless signal. |                                                                                                                                             |
| 80H~86H | Read  | PID     | Product information 7 bytes |                                                                                                                                             |   |           | 7 bytes of product information                                                                                                                                                                                                   |                                                                                                                                             |

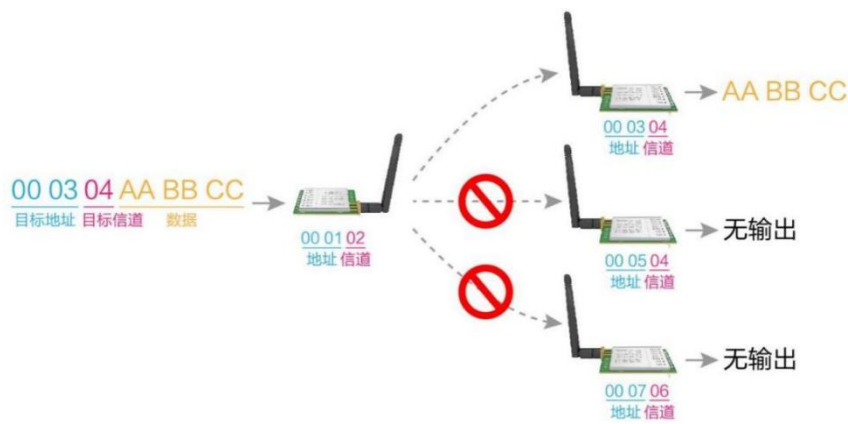
#### 4.4.3 LoRa factory default parameters

|              |                                                                                                                                                                                     |         |         |           |      |               |                    |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|-----------|------|---------------|--------------------|
| Model number | EWM22A-400BWL22S factory default parameter values: C0 00 09 00 00 00 62 00 17 03 00 00 00<br>EWM22A-900BWL22S factory default parameter values: C0 00 09 00 00 00 62 00 12 03 00 00 |         |         |           |      |               |                    |
| Model number | Frequency                                                                                                                                                                           | Address | Channel | Air speed | Baud | Serial Format | Transmission power |





|                      |                |        |      |         |        |     |       |
|----------------------|----------------|--------|------|---------|--------|-----|-------|
| EWM22A-400BW<br>L22S | 433.125MH<br>z | 0x0000 | 0x17 | 2.4kbps | 115200 | 8N1 | 22dbm |
| EWM22A-900BW<br>L22S | 868.125MH<br>z | 0x0000 | 0x12 | 2.4kbps | 115200 | 8N1 | 22dbm |



#### 4.4.4 LoRa Transmit, Receive Instructions

##### 1.fixed point transmitter

##### 2. Broadcast transmitter

##### 3. Broadcast Address

Example: set module A address to 0xFFFF and channel to 0x04.

When module A acts as a transmitter (same mode, transparent transmission method), all receiving modules under channel 0x04 can receive the data for the purpose of broadcastin

##### 4. Listening address

Example: Set module A address to 0xFFFF and channel to 0x04.

When module A is used as a receiver, it can receive all the data under channel 0x04 to achieve the purpose of listening.

##### 5. AUX detailed explanation

AUX is used for LoRa transceiver buffer indication and self-test indication.

It indicates whether the module has data that has not yet been emitted through LoRa, or whether the wireless data that has been received has not yet been sent out all through the serial port, or whether the module is in the process of initializing self-test.

## 5 AT command

### 5.1 Command format

| Command format | Description     | Example        |
|----------------|-----------------|----------------|
| AT+XXX         | Run command     | AT+RESET       |
| AT+XXX=?       | Query command   | AT+HELP=?      |
| AT+XXX=YYYY    | Setting command | AT+BLENAME=ABC |

- Module supports parameter reading and writing, the default baud rate supported by the module serial port is 115200bps, 8N1 format.
- UART, BLE, and TCP commands do not bring back carriage returns;
- The return result of UART configuration mode ends with (\r) and (\n), BLE configuration return value does not end with (\r) and (\n);
- Serial port baud rate defaults to 115200, 8bit data bit, 1 stop bit, no parity;
- Instruction parameters are in ASCII format;
- Instructions are case-sensitive.
- Some instructions do not support UART configuration, such as AT+AUTH, AT+AUTHEXIT;
- All AT instructions cannot contain invisible characters such as spaces and tabs.

### 5.2 Return Value Description

| AT instruction return value | Explanation                                                       |
|-----------------------------|-------------------------------------------------------------------|
| AT_OK                       | Correct                                                           |
| AT_PARAM_ERROR              | Setting parameter error                                           |
| AT_NVS_ERROR                | NVS error                                                         |
| AT_AUTH_ERR                 | Permission error, need to use AT+AUTH to configure authentication |
| AT_ERROR                    | Other errors                                                      |

## 5.3 Status output

| Status output                        | Description                         |
|--------------------------------------|-------------------------------------|
| BLE CONNECT                          | BLE connection successful           |
| BLE DISCONNECT                       | BLE disconnected                    |
| WIFI START CONNECT:ssid,password     | WIFI start to connect               |
| CONNECTED AP GOT IP: 192.168.x.x     | WIFI connection successful          |
| RECONNECT:ssid,password              | WIFI reconnect                      |
| TCP CLIENT START                     | TCP client started to connect       |
| UNABLE TO CONNECT:192.168.xx.xx:3333 | TCP client cannot connect to server |
| TCP CONNECTED                        | TCP client connects to server       |
| TCP DISCONNECTED                     | TCP client disconnects              |

The above states can be turned on or off by the AT command “AT+ LOGMSG”, see the AT command section for details.

## 5.4 AT command table

| Categories    | Command       | Description                         |
|---------------|---------------|-------------------------------------|
| Basic Command | AT            | AT Link Check                       |
|               | AT+FWCODE     | Query Firmware Version              |
|               | AT+UID        | Query MAC                           |
|               | AT+DEVTYPE    | Query Product Model                 |
|               | AT+RESET      | Reset                               |
|               | AT+DEFAULT    | Restore Factory Settings            |
|               | AT+UART       | Setting/Querying Baud Rate          |
|               | AT+HELP       | Query all commands                  |
|               | AT+HMODE      | Query/set operating mode            |
|               | AT+SMODE      | Query/set listening function        |
|               | AT+AUTHKEY    | Query/Set BLE/WIFI pairing key      |
|               | AT+AUTH       | BLE/WIFI authentication pairing key |
|               | AT+AUTHEXIT   | Exit key verification               |
|               | AT+LIGHTSLEEP | Enter shallow sleep                 |
|               | AT+DEEPSLEEP  | Enter deep sleep                    |
|               | AT+SLEEPKEEP  | Deep Hibernation Hold Function      |
|               | AT+COMSW      | BLE/WIFI/LORA forced switching      |
|               | ATE0          | Turn off display back               |
|               | ATE1          | Open display                        |
|               | AT+LOGMSG     | Status Print                        |

|                     |             |                                                |
|---------------------|-------------|------------------------------------------------|
| WIFI/TCP<br>Command | AT+STA      | Setting/querying connected WIFI settings       |
|                     | AT+CIPSTATE | Setting/querying TCP connection IP and port    |
| 2.4G Command        | AT+NOWADDR  | Setting/querying the 2.4G transmit destination |
|                     | AT+NOWPER   | Set/Query 2.4G Prefix Switch                   |
| BLE Command         | AT+BLENAM   | Set/Query BLE Name                             |
|                     | AT+BLEADV   | Set/query BLE broadcast interval               |
|                     | AT+BLEOTA   | Set/Query BLE upgrade function                 |
|                     | AT+BLEPOWER | Set/Query BLE transmit power                   |
|                     | AT+RATE     | Set/Query Air Rate                             |
|                     | AT+PACKET   | Set/Query Packet Length                        |
|                     | AT+WOR      | Set/Query WOR Role                             |
|                     | AT+WTIME    | Set/Query WOR period                           |
|                     | AT+POWER    | Set/Query Transmit Power                       |
|                     | AT+TRANS    | Set/Query Transmit Mode                        |
|                     | AT+ROUTER   | Set/query relay mode                           |
|                     | AT+LBT      | Query Listen Before Talk switch                |
|                     | AT+ERSSI    | Query ambient noise RSSI switch                |
|                     | AT+DRSSI    | Query RSSI output                              |
|                     | AT+ADDR     | Query module address                           |
|                     | AT+CHANNEL  | Query module working channel                   |
|                     | AT+NETID    | Set Network ID                                 |
|                     | AT+KEY      | Setting the module key                         |

## 5.5 Description of the AT command

### 5.5.1 AT link check

| Command | Responsive | Example                                                    |
|---------|------------|------------------------------------------------------------|
| AT      | AT_OK      | Send: AT<br>Return: AT_OK //command successfully responded |

### 5.5.2 Check Firmware Version

| Command     | Responsive       | Example           |
|-------------|------------------|-------------------|
| AT+FWCODE=? | <version number> | Send: AT+FWCODE=? |

|  |       |                                   |
|--|-------|-----------------------------------|
|  | AT_OK | Return: FWCODE=75XX-0-10<br>AT_OK |
|--|-------|-----------------------------------|

### 5.5.3 Check UID

| Command                                                                                                                                                             | Responsive                                                  | Example                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| AT+UID=?                                                                                                                                                            | < BASE MAC><br>AT_OK                                        | Send: AT+UID=?<br>Return: d0:ef:76:9b:3e:34<br>AT_OK //command successfully responded |
| < base MAC >                                                                                                                                                        | The returned BASE MAC address is in hexadecimal characters. |                                                                                       |
| Description:<br><BASE MAC> chip base MAC, <BLE MAC> adds 2 to this base. e.g. a module with a <BASE MAC> of d0:ef:76:9b:3e:34 has a <BLE MAC> of d0:ef:76:9b:3e:36. |                                                             |                                                                                       |

### 5.5.4 Module Model

| Command        | Responsive                                              | Example                                                                                          |
|----------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| AT+DEVTYPE=?   | <Module Model><br>AT_OK                                 | Send: AT+DEVTYPE=?<br>Return: DEVTYPE=EWM22A-900BWL22S<br>AT_OK //command successfully responded |
| <Module Model> | Return Module Model:EWM22A-900BWL22S, EWM22A-400BWL22S; |                                                                                                  |

### 5.5.5 Reset Chips

| Command  | Responsive | Example                         |
|----------|------------|---------------------------------|
| AT+RESET | AT_OK      | Send: AT+RESET<br>Return: AT_OK |

### 5.5.6 Restore Factory Setting

| Command                                     | Responsive | Example                                                            |
|---------------------------------------------|------------|--------------------------------------------------------------------|
| AT+DEFAULT                                  | AT_OK      | Send: AT+DEFAULT<br>Return: AT_OK //command successfully responded |
| Description:<br>Will restart automatically. |            |                                                                    |

### 5.5.7 Serial port setting

| Command                                                                                                                                                                                                                                                                                                                                                                                                                           | Responsive                                                                                                        | Example                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| AT+UART=?                                                                                                                                                                                                                                                                                                                                                                                                                         | AT_OK                                                                                                             | Send: AT+UART=?<br>Return: 7,0<br>AT_OK //command successfully responded   |
| AT+UART=<Baud rate>,<Check digit>.                                                                                                                                                                                                                                                                                                                                                                                                | AT_OK                                                                                                             | Send: AT+UART=3,0<br>Return: 3,0<br>AT_OK //command successfully responded |
| <Baud rate>                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 --- 1200<br>1 --- 2400<br>2 --- 4800<br>3 --- 9600<br>4 --- 19200<br>5 --- 38400<br>6 --- 57600<br>7 --- 115200 |                                                                            |
| <check digit>                                                                                                                                                                                                                                                                                                                                                                                                                     | 0 --- disable<br>1 --- odd<br>2 --- even<br>3 --- disable                                                         |                                                                            |
| Description:<br>Reboot to take effect, power down to save;<br>Two modules communicating with each other can have different serial port baud rates and different checksums;<br>When transmitting larger data packets continuously, users need to consider the data blocking brought by the same baud rate, and may even be lost;<br>It is generally recommended that the baud rates of the two communicating parties are the same. |                                                                                                                   |                                                                            |

### 5.5.8 AT Instruction Query

| Command   | Responsive             | Example                                                           |
|-----------|------------------------|-------------------------------------------------------------------|
| AT+HELP=? | <command list<br>AT_OK | Send: AT+HELP=?<br>Return: AT_OK //command successfully responded |

### 5.5.9 AT Disable display back

| Command | Responsive | Example                                                      |
|---------|------------|--------------------------------------------------------------|
| ATE0    | AT_OK      | Send: ATE0<br>Return: AT_OK //command successfully responded |

### 5.5.10 AT Enable display back

| Command | Responsive | Example                                                      |
|---------|------------|--------------------------------------------------------------|
| ATE1    | AT_OK      | Send: ATE1<br>Return: AT_OK //command successfully responded |

### 5.5.11 Status Printing

| Command                                                    | Responsive                      | Example                                                                   |
|------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------|
| AT+LOGMSG=?                                                | <Status Print Switch><br>AT_OK  | Send: AT+LOGMSG=?<br>Return: 11<br>AT_OK //command successfully responded |
| AT+LOGMSG=<output switch>                                  | <output switch><br>AT_OK        | Send: AT+LOGMSG=1<br>Return: 1<br>AT_OK //command successfully responded  |
| <Status Print Switch>                                      | 0 --- off<br>1 --- On (default) |                                                                           |
| Description:<br>Effective immediately, power down to save; |                                 |                                                                           |

### 5.5.12 BLE/WIFI/LORA Forced switch

| Command                                                                                                                                                                   | Responsive                                      | Example                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|
| AT+COMSW=?                                                                                                                                                                | L: <Switch>,W:<br><Switch>,B: <Switch><br>AT_OK | Send: AT+COMSW=?<br>Return: L:1,W:1,B:1<br>AT_OK //command successfully responded |
| AT+COMSW=<switch>,<switch>,<switch>,<switch>                                                                                                                              | <Switch<br>AT_OK                                | Send: AT+COMSW=1,1,1<br>Return: 1<br>AT_OK //command successfully responded       |
| <Switch>                                                                                                                                                                  | 0 --- off<br>1 --- On (default)                 |                                                                                   |
| Description:<br>Reboot to take effect, power down to save;<br>Force off BLE/WIFI/LORA, use this command with caution, it may conflict with the function of various modes. |                                                 |                                                                                   |

### 5.5.13 BLE/WIFI key pairing

| Command                                                    | Responsive                                                                                                      | Example                                                                             |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| AT+AUTHKEY=?                                               | <AUTHKEY><br>AT_OK                                                                                              | Send: AT+AUTHKEY=?<br>123456<br>Return: AT_OK //command successfully responded      |
| AT+ AUTHKEY=XXXXXX                                         | <AUTHKEY><br>AT_OK                                                                                              | Send: AT+AUTHKEY=123456<br>123456<br>Return: AT_OK //command successfully responded |
| <AUTHKEY>                                                  | BLE, TCP into the configuration mode need to verify the key, the length of 6-15 characters, the default 123456. |                                                                                     |
| Description:<br>Effective immediately, power down to save; |                                                                                                                 |                                                                                     |

### 5.5.14 BLE/WIFI Configuring Authentication

| Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Responsive | Example                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------|
| AT+AUTH=XXXXXX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AT_OK      | Send: AT+AUTH=123456<br>Return: AT_OK //command successfully responded |
| Description:<br>Valid for single connection, need to authenticate again after reconnecting ;<br>After BLE, TCP and the module are connected, you can send this command through the configuration channel, and after the return success, you can use all AT commands through the configuration channel;<br>If there is no authorization to send commands directly, the module returns “AT_AUTH_ERR”.<br>This command is only used for BLE configuration, UART can not use this command, to modify the key use AT+AUTHKEY command. |            |                                                                        |

### 5.5.15 Exit Configuration Authentication

| Command                                                                                                                                                                                                                                                              | Responsive | Example                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------|
| AT+AUTHEXIT                                                                                                                                                                                                                                                          | AT_OK      | Send: AT+AUTHEXIT<br>Return: AT_OK //command successfully responded |
| Description:<br>BLE, TCP and module connection, through the AT + AUTH = XXXXXXXXX into the configuration of the successful authentication before using this configuration;<br>This command is only used for BLE and TCP configuration, UART cannot use this command. |            |                                                                     |



### 5.5.16 Shallow sleep

| Command                                                                                                                                                                   | Responsive | Example                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------|
| AT+LIGHTSLEEP                                                                                                                                                             | AT_OK      | Send: AT+LIGHTSLEEP<br>Return: AT_OK |
| Description:<br>Turn off WIFI, BLE, LoRa to significantly reduce power consumption;<br>Can use serial port to wake up, power consumption is higher than deep hibernation; |            |                                      |

### 5.5.17 Deep sleep

| Command                                                                                                                                                        | Responsive | Example                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------|
| AT+DEEPSLEEP                                                                                                                                                   | AT_OK      | Send: AT+DEEPSLEEP<br>Return: AT_OK //command successfully responded |
| Description:<br>Turning off WIFI, BLE, LoRa significantly reduces power consumption;<br>Wake-up via wake pin low consumes less power than shallow hibernation. |            |                                                                      |

### 5.5.18 Working mode

| Command                                                                        | Responsive                                                                                                                                                                                                                                                                                                                                                                   | Example                                                                 |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| AT+HMODE=?                                                                     | <working mode><br>AT_OK                                                                                                                                                                                                                                                                                                                                                      | Send: AT+HMODE=?<br>Return: 0<br>AT_OK //command successfully responded |
| AT+HMODE=<working mode>                                                        | AT_OK                                                                                                                                                                                                                                                                                                                                                                        | Send: AT+HMODE=3<br>Return: 3<br>AT_OK //command successfully responded |
| <working mode>                                                                 | 0 --- Configuration mode<br>1 --- UART/LORA pass-through, BLE listening<br>2 --- UART/BLE passthrough, LORA listening<br>3 --- LORA/BLE passthrough, UART listening<br>4 --- UART/LORA passthrough, WIFI listening<br>5 --- LORA/WIFI passthrough, UART listening<br>6 --- UART/WIFI passthrough, LORA listening<br>7 --- LORA/UART passthrough, WOR mode, WIFI/BLE both off |                                                                         |
| Description:<br>After switching the module, the module restarts automatically; |                                                                                                                                                                                                                                                                                                                                                                              |                                                                         |

### 5.5.19 Deep sleep hold function

| Command                                                                                                                                                                                               | Responsive                      | Example                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------|
| AT+SLEEPKEEP=?                                                                                                                                                                                        | <Hold switch><br>AT_OK          | Send: AT+SLEEPKEEP=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+SLEEPKEEP=<Hold switch>                                                                                                                                                                            | AT_OK                           | Send: AT+SLEEPKEEP=1<br>Return: 1<br>AT_OK //command successfully responded |
| <Hold switch>                                                                                                                                                                                         | 0 --- off (default)<br>1 --- On |                                                                             |
| Description:<br>Effective immediately, power down save;<br>If this function is turned on, the module still enters deep hibernation after power down and restart, and can be woken up by wake pin low. |                                 |                                                                             |

### 5.5.20 Listening Settings

| Command                                                                                                                                                               | Responsive                                  | Example                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------|
| AT+SMODE=?                                                                                                                                                            | <monitor switch>,<listener prefix><br>AT_OK | Send: AT+SMODE=?<br>Return: 1,1<br>AT_OK //command successfully responded   |
| AT+SMODE=1,1                                                                                                                                                          | AT_OK                                       | Send: AT+SMODE=1.1<br>Return: 1,1<br>AT_OK //command successfully responded |
| <Listening switch>                                                                                                                                                    | 0 --- off<br>1 --- On (default)             |                                                                             |
| <listener prefix>                                                                                                                                                     | 0 --- off<br>1 --- On (default)             |                                                                             |
| Description:<br>Effective immediately, power-down save;<br>Controls whether the listening end listens for data and whether it prints the data flow and length prefix. |                                             |                                                                             |

### 5.5.21 WIFI STA

| Command                  | Responsive                 | Example                                                                                      |
|--------------------------|----------------------------|----------------------------------------------------------------------------------------------|
| AT+STA=?                 | <SSID>,<PASSWORD><br>AT_OK | Send: AT+STA=?<br>Return: KingOfSoftware,123456789<br>AT_OK //command successfully responded |
| AT+STA=wifissid,wifipswd | <SSID>,<PASSWORD>          | Send: AT+STA=wifissid,wifipswd                                                               |

|                                                                                                                                                       |                                                       |                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------|
|                                                                                                                                                       | AT_OK                                                 | Return: wifissid,wifipswd<br>AT_OK //command successfully responded |
| <SSID>                                                                                                                                                | SSID of the target AP. length 1-32 characters.        |                                                                     |
| <PASSWORD>                                                                                                                                            | Password of the target AP. Length is 8-32 characters. |                                                                     |
| Description:<br>Reboot to take effect, power down to save;<br>Can only connect to 2.4G AP, and only support WPA2 WIFI security authentication method. |                                                       |                                                                     |

## 5.5.22 Connection target TCP server IP/PORT

| Command                                                                                             | Responsive                      | Example                                                                                                        |
|-----------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| AT+CIPSTATE=?                                                                                       | <TCP server IP>,<PORT><br>AT_OK | Send: AT+CIPSTATE=?<br>Return: 192.168.31.102,3333<br>AT_OK //command successfully responded                   |
| AT+CIPSTATE=192.168.31.102,3333                                                                     | <TCP server IP>,<PORT><br>AT_OK | Send: AT+CIPSTATE=192.168.31.102,3333<br>Return: 192.168.31.102,3333<br>AT_OK //command successfully responded |
| Description:<br>Reboot to take effect, power down to save.<br>Only IPV4 IP addresses are supported. |                                 |                                                                                                                |

## 5.5.23 BLE broadcast name

| Command                                                    | Responsive                                                                                                                          | Example                                                                                  |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| AT+BLENAME=?                                               | <BLE broadcast name><br>AT_OK                                                                                                       | Send: AT+BLENAME=?<br>Return: EWM22A-900BWL22S<br>AT_OK //command successfully responded |
| AT+BLENAME=<BLE broadcast name>                            | <BLE broadcast name><br>AT_OK                                                                                                       | Send: AT+BLENAME=ABC<br>Return: ABC<br>AT_OK //command successfully responded            |
| <BLE broadcast name>                                       | Broadcast name length no greater than 16 bytes; factory default broadcast name is module model (EWM22A-900BWL22S, EWM22A-400BWL22S) |                                                                                          |
| Description:<br>Reboot to take effect, power down to save; |                                                                                                                                     |                                                                                          |

## 5.5.24 BLE Broadcast Intervals

| Command     | Responsive           | Example           |
|-------------|----------------------|-------------------|
| AT+BLEADV=? | <broadcast interval> | Send: AT+BLEADV=? |

|                                                                                                                                                                 |                                                                                       |                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
|                                                                                                                                                                 | AT_OK                                                                                 | Return: 100<br>AT_OK //command successfully responded                        |
| AT+BLEADV=<broadcast interval>                                                                                                                                  | <broadcast interval><br>AT_OK                                                         | Send: AT+BLEADV=100<br>Return: 100<br>AT_OK //command successfully responded |
| <broadcast interval>                                                                                                                                            | Broadcast interval, parameter value range 32-16384, default 1600, 1600*0.625ms=1000ms |                                                                              |
| Description:<br>Reboot to take effect, power down to save<br>The larger the broadcast interval, the longer the update time and the lower the power consumption; |                                                                                       |                                                                              |

### 5.5.25 BLE power setting

| Command                                                                                                                                                                                      | Responsive                                                                                                      | Example                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| AT+BLEPOWER=?                                                                                                                                                                                | <BLE power level><br>AT_OK                                                                                      | Send: AT+BLEOTA=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+BLEOTA =<BLE power level>                                                                                                                                                                 | <BLE power level><br>AT_OK                                                                                      | Send: AT+BLEOTA=1<br>Return: 1<br>AT_OK //command successfully responded |
| <BLE power level>                                                                                                                                                                            | 0 --- +20dbm<br>1 --- +15dbm<br>2 --- +9dbm<br>3 --- +3dbm<br>4 --- 0dbm(default)<br>5 --- -3dbm<br>6 --- -9dbm |                                                                          |
| Description:<br>Power and current are non-linearly related, and the power supply is most efficient at maximum power;<br>Current does not decrease in the same proportion as power decreases. |                                                                                                                 |                                                                          |

### 5.5.26 BLE Upgrade settings

| Command                         | Responsive                    | Example                                                                   |
|---------------------------------|-------------------------------|---------------------------------------------------------------------------|
| AT+BLEOTA=?                     | <BLE Upgrade Switch><br>AT_OK | Send: AT+BLEOTA=?<br>Return: 1<br>AT_OK //command successfully responded  |
| AT+BLEOTA =<BLE Upgrade Switch> | <BLE Upgrade Switch><br>AT_OK | Send: AT+BLEOTA =1<br>Return: 1<br>AT_OK //command successfully responded |
| <BLE Upgrade Switch>            | 0 --- off (default)           |                                                                           |

|                                                                                                                                                                                                                                                                                                                                                                                          |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                          | 1 --- On |
| <b>Description</b><br>Immediately effective, power down to save.<br>BLE OTA will occupy a lot of stack, enable this function will only run BLE function to avoid upgrade failure.<br>The module will restart automatically after successful upgrade, firmware upgrade may add settings, it is recommended to manually restore factory settings and reconfigure the module after upgrade; |          |

### 5.5.27 LoRa airspeed

| Command                                                                                                                                                                                   | Responsive                                                                                                      | Example                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| AT+RATE=?                                                                                                                                                                                 | <airspeed><br>AT_OK                                                                                             | <b>Send:</b> AT+RATE=?<br><b>Return:</b> 2<br><b>AT_OK //command successfully responded</b> |
| AT+RATE=<airspeed>                                                                                                                                                                        | <airspeed><br>AT_OK                                                                                             | <b>Send:</b> AT+RATE=1<br><b>Return:</b> 1<br><b>AT_OK //command successfully responded</b> |
| <airspeed>                                                                                                                                                                                | 0 --- 2.4K<br>1 --- 2.4K<br>2 --- 2.4K<br>3 --- 4.8K<br>4 --- 9.6K<br>5 --- 19.2K<br>6 --- 38.4K<br>7 --- 62.5K |                                                                                             |
| <b>Description:</b><br>The air rate must be the same for both sides of the LoRa communication;<br>The higher the air rate, the lower the delay and the shorter the transmission distance. |                                                                                                                 |                                                                                             |

### 5.5.28 LoRa Packet length

| Command                   | Responsive               | Example                                                                                       |
|---------------------------|--------------------------|-----------------------------------------------------------------------------------------------|
| AT+PACKET=?               | <Packet length><br>AT_OK | <b>Send:</b> AT+PACKET=?<br><b>Return:</b> 1<br><b>AT_OK //command successfully responded</b> |
| AT+PACKET=<Packet length> | <Packet length><br>AT_OK | <b>Send:</b> AT+PACKET=1<br><b>Return:</b> 1<br><b>AT_OK //command successfully responded</b> |
| <Packet length>           | 0 --- 240 (default)      |                                                                                               |

|                                                                                                                                                                                                                                                                                              |                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                                                                                                                                                                                                                                                                                              | 1 --- 128<br>2 --- 64<br>3 --- 32 |
| Description:<br>The user sends data less than the packet length, the serial port output at the receiving end is presented as uninterrupted continuous output;<br>If the data sent by the user is larger than the packet length, the serial port at the receiving end will output in packets. |                                   |

### 5.5.29 LoRa WOR role

| Command                                | Responsive                                                                                                                                                                                                                                                                                           | Example                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| AT+WOR=?                               | <WOR role ><br>AT_OK                                                                                                                                                                                                                                                                                 | Send: AT+PACKET=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+WOR =< WOR role>                    | < WOR role ><br>AT_OK                                                                                                                                                                                                                                                                                | Send: AT+PACKET=1<br>Return: 1<br>AT_OK //command successfully responded |
| <WOR role>                             | 0 --- WOR receiver (default)<br>Working in WOR listening mode with the listening period described below (WOR period) saves a lot of power consumption.<br><br>1 --- WOR transmitter<br>Module transmits and receives on and adds a wake-up code for a certain period of time when transmitting data. |                                                                          |
| Description:<br>Valid only for mode 7; |                                                                                                                                                                                                                                                                                                      |                                                                          |

### 5.5.30 LoRa WOR cycle

| Command               | Responsive                                                                  | Example                                                                 |
|-----------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| AT+WTIME=?            | <time index><br>AT_OK                                                       | Send: AT+WTIME=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+WTIME=<time index> | <time index><br>AT_OK                                                       | Send: AT+WTIME=1<br>Return: 1<br>AT_OK //command successfully responded |
| <time index>          | 0 --- 500ms<br>1 --- 1000ms<br>2 --- 1500ms<br>3 --- 2000ms<br>4 --- 2500ms |                                                                         |

|                                                                                                                                                                                                                                                                                                                                             |                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                             | 5 --- 3000ms<br>6 --- 3500ms<br>7 --- 4000ms (default) |
| Description:<br>Valid only for mode 7;<br>Cycle time $T = (1 + \text{WOR}) * 500\text{ms}$ , the maximum is 4000ms and the minimum is 500ms;<br>The longer the period time between WOR listening intervals, the lower the average power consumption, but the higher the data delay;<br>Both sender and receiver must agree (very important) |                                                        |

### 5.5.31 LoRa firing power

| Command                                                                                                                                                                      | Responsive                                               | Example                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------|
| AT+POWER=?                                                                                                                                                                   | <Power Index><br>AT_OK                                   | Send: AT+WTIME=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+POWER=<Power Index>                                                                                                                                                       | <Power Index><br>AT_OK                                   | Send: AT+WTIME=1<br>Return: 1<br>AT_OK //command successfully responded |
| <Power Index>                                                                                                                                                                | 0 --- 22dBm<br>1 --- 22dBm<br>2 --- 22dBm<br>3 --- 22dBm |                                                                         |
| Power and current are non-linearly related, and the power supply is most efficient at maximum power;<br>Current does not decrease in the same proportion as power decreases. |                                                          |                                                                         |

### 5.5.32 LoRa transfer mode

| Command                                                                                                                                                                               | Responsive                                       | Example                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------|
| AT+TRANS=?                                                                                                                                                                            | <LoRa transfer mode><br>AT_OK                    | Send: AT+TRANS=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+TRANS=<LoRa transfer mode>                                                                                                                                                         | <LoRa transfer mode><br>AT_OK                    | Send: AT+TRANS=1<br>Return: 1<br>AT_OK //command successfully responded |
| <LoRa transfer mode>                                                                                                                                                                  | 0 --- transparent (default)<br>1 --- fixed point |                                                                         |
| For fixed-point transmission, the module recognizes the first three bytes of the serial data as: address high + address low + channel, and uses them as the wireless transmit target. |                                                  |                                                                         |

### 5.5.33 LoRa relay mode

| Command                                                                                                                                                                                                                                                                                                                   | Responsive                      | Example                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------|
| AT+ROUTER=?                                                                                                                                                                                                                                                                                                               | <LoRa relay switch><br>AT_OK    | <b>Send: AT+ROUTER=?</b><br><b>Return: 1</b><br><b>AT_OK //command successfully responded</b> |
| AT+ROUTER=<LoRa relay switch>                                                                                                                                                                                                                                                                                             | <LoRa relay switch><br>AT_OK    | <b>Send: AT+ROUTER=1</b><br><b>Return: 1</b><br><b>AT_OK //command successfully responded</b> |
| <LoRa relay switch>                                                                                                                                                                                                                                                                                                       | 0 --- off (default)<br>1 --- On |                                                                                               |
| <p>When the relay function is enabled, the module will initiate a forwarding if the destination address is not the module itself;</p> <p>In order to prevent data backhauling, it is recommended to be used in conjunction with fixed-point mode; i.e., the destination address is different from the source address.</p> |                                 |                                                                                               |

### 5.5.34 LoRa Listen Before Talk

| Command                                                                                                                                                                                                                                                              | Responsive                      | Example                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------|
| T+LBT=?                                                                                                                                                                                                                                                              | <LBT switch><br>AT_OK           | <b>Send: T+LBT=?</b><br><b>Return: 1</b><br><b>AT_OK //command successfully responded</b> |
| T+LBT=<LBT switch>                                                                                                                                                                                                                                                   | <LBT switch><br>AT_OK           | <b>Send: T+LBT=1</b><br><b>Return: 1</b><br><b>AT_OK //command successfully responded</b> |
| <LBT switch>                                                                                                                                                                                                                                                         | 0 --- off (default)<br>1 --- On |                                                                                           |
| <p>When enabled, wireless data will be listened to before transmitting, which can avoid interference to some extent, but may bring data delay;</p> <p>The maximum dwell time of LBT is 2 seconds, and it will be forced to send out when it reaches two seconds.</p> |                                 |                                                                                           |

### 5.5.35 LoRa Environmental RSSI (erssi)

| Command               | Responsive               | Example                                                                                      |
|-----------------------|--------------------------|----------------------------------------------------------------------------------------------|
| AT+ERSSI=?            | < ERSSI switch><br>AT_OK | <b>Send: AT+ERSSI=?</b><br><b>Return: 1</b><br><b>AT_OK //command successfully responded</b> |
| T+LBT=< ERSSI switch> | < ERSSI switch><br>AT_OK | <b>Send: AT+ERSSI=1</b><br><b>Return: 1</b>                                                  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 |                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 | AT_OK //command successfully responded |
| < ERSSI switch>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0 --- off (default)<br>1 --- On |                                        |
| Description:<br>Reboot to take effect, power down to save;<br>When enabled, you can send instruction C0 C1 C2 C3 instruction to read registers in UART-IoRa mode or WOR transmit mode;<br>Register 0x00 : current ambient noise RSSI;<br>Register 0X01 : RSSI of the last received data.<br>(The current channel noise is: dBm = -(256 - RSSI));<br>Instruction format: C0 C1 C2 C3 + start address + read length;<br>return: C1 + address + read length + read the effective value; for example: send C0 C1 C2 C3 00 01<br>Return C1 00 01 RSSI (address can only start from 00) |                                 |                                        |

### 5.5.36 LoRa Data RSSI (data\_rssi)

| Command                                                                                                                                            | Responsive                      | Example                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------|
| AT+DRSSI=?                                                                                                                                         | < DRSSI switch<br>AT_OK         | Send: AT+DRSSI=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+DRSSI=< DRSSI switch>                                                                                                                           | < DRSSI switch<br>AT_OK         | Send: T+DRSSI=1<br>Return: 1<br>AT_OK //command successfully responded  |
| < DRSSI switch>                                                                                                                                    | 0 --- off (default)<br>1 --- On |                                                                         |
| Description:<br>When enabled, the module receives wireless data, which will follow an RSSI intensity byte when output through the serial port TXD. |                                 |                                                                         |

### 5.5.37 LoRa module address

| Command                                                | Responsive                                  | Example                                                                |
|--------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------|
| AT+ADDR=?                                              | <module address><br>AT_OK                   | Send: AT+ADDR=?<br>Return: 1<br>AT_OK //command successfully responded |
| AT+ADDR=<module address>                               | <module address><br>AT_OK                   | Send: AT+ADDR=1<br>Return: 1<br>AT_OK //command successfully responded |
| <module address>                                       | Module address 0~65535 (decimal), default 0 |                                                                        |
| Description:<br>Module address high byte and low byte; |                                             |                                                                        |

Note: When the module address is equal to FFFF, it can be used as the broadcast and listening address, i.e.: at this point the module will not be address filtered

### 5.5.38 LoRa module channel

| Command                                                                                                                           | Responsive                                                                                                                                                     | Example                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| AT+CHANNEL=?                                                                                                                      | < CH ><br>AT_OK                                                                                                                                                | Send: AT+CHANNEL=?<br>Return: 18<br>AT_OK //command successfully responded   |
| AT+CHANNEL=< CH >                                                                                                                 | <module channel><br>AT_OK                                                                                                                                      | Send: AT+ CHANNEL=18<br>Return: 18<br>AT_OK //command successfully responded |
| < CH >                                                                                                                            | Channel Control (CH)<br>0-83 represent a total of 84 channels (for 400 band) respectively<br>0-80 represent a total of 81 channels respectively (for 900 band) |                                                                              |
| Description: Actual frequency = 410.125 + CH * 1M<br>Actual frequency = 410.125 + CH * 1M<br>Actual frequency = 850.125 + CH * 1M |                                                                                                                                                                |                                                                              |

### 5.5.39 LoRa Network ID

| Command                                                                                                                             | Responsive                                | Example                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------|
| AT+NETID=?                                                                                                                          | <Network ID ><br>AT_OK                    | Send: AT+NETID=?<br>Return: 0<br>AT_OK //command successfully responded |
| AT+NETID=<Network ID >                                                                                                              | <Network ID ><br>AT_OK                    | Send: AT+NETID=0<br>Return: 0<br>AT_OK //command successfully responded |
| <Network ID >                                                                                                                       | Module network 0~255 (decimal), default 0 |                                                                         |
| Description:<br>Network address for distinguishing networks;<br>When communicating with each other, they should be set to the same. |                                           |                                                                         |

### 5.5.40 LoRa Key (key)

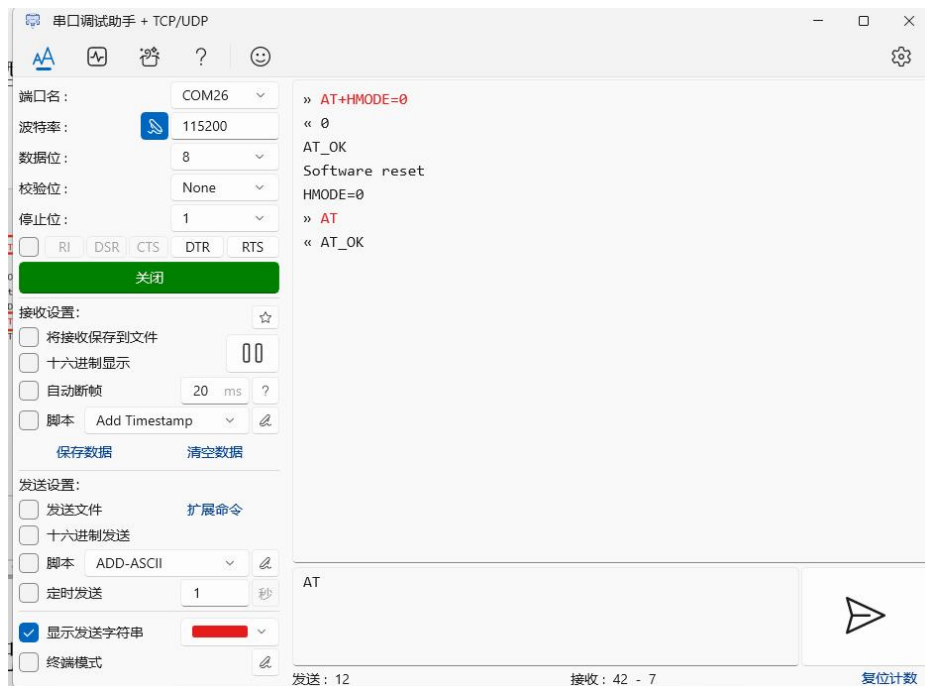
| Command  | Responsive     | Example                                                               |
|----------|----------------|-----------------------------------------------------------------------|
| AT+KEY=? | <KEY><br>AT_OK | Send: AT+KEY=?<br>Return: 0<br>AT_OK //command successfully responded |

|                                                                                                                                                                                                                                                                                                    |                                        |                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------|
| AT+KEY=<KEY>                                                                                                                                                                                                                                                                                       | <KEY><br>AT_OK                         | Send: AT+KEY=1<br>Return: 0<br>AT_OK //command successfully responded |
| <KEY>                                                                                                                                                                                                                                                                                              | Module key 0~65535 (decimal),default 0 |                                                                       |
| <p>Description:</p> <p>Write only, read returns 0;</p> <p>Used for encryption to avoid interception of over-the-air wireless data by similar modules;</p> <p>The module will internally use these two bytes as a calculation factor to transform and encrypt the over-the-air wireless signal.</p> |                                        |                                                                       |

## 6 Quick Start

### 1. UART AT command configuration module

AT+HMODE=0 //Enter the configuration module

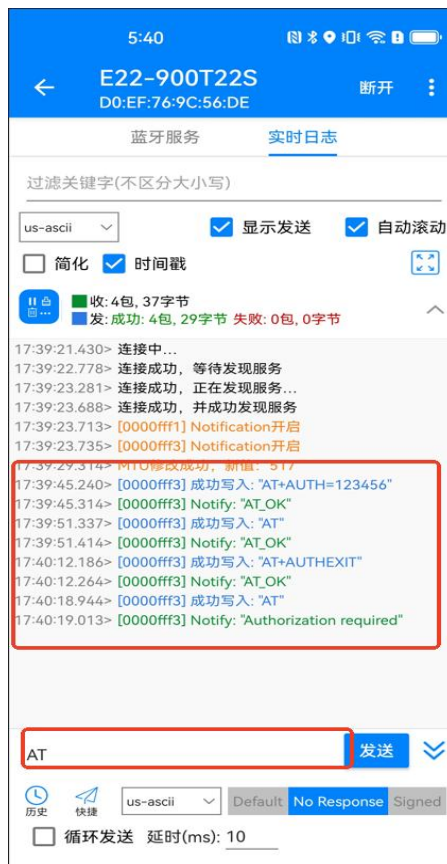
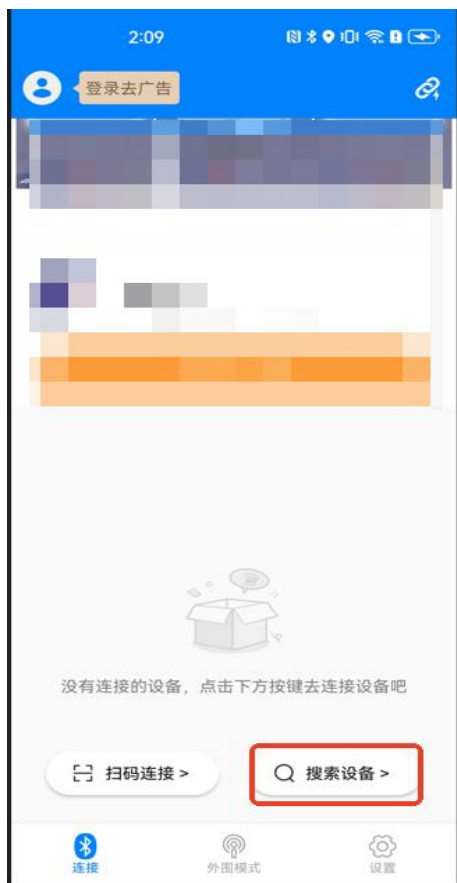


### 2. BLE AT command configuration module

#### 1、Preparation of cell phone APP (Android phone)



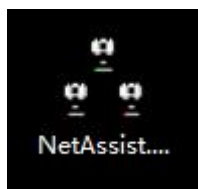
#### 2、Connect to the module Bluetooth, send AT+AUTH=123456



### 3. TCP AT command configuration module

#### 1、Preparation

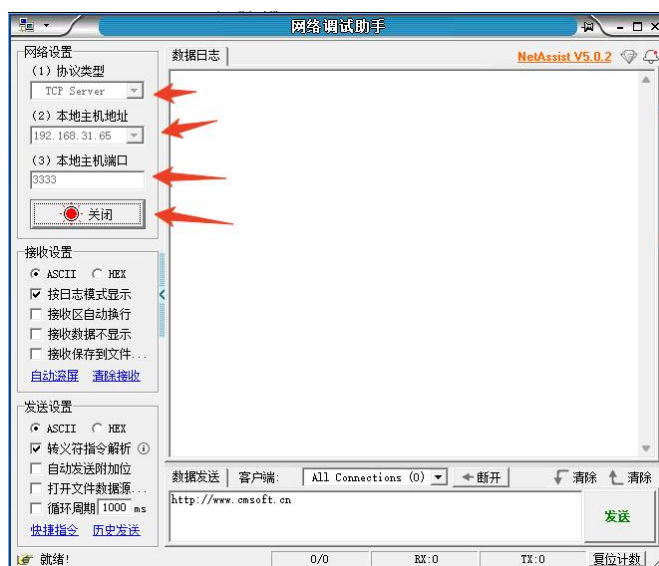
##### ① NetAssist.exe



② A computer, and ensure that the module and the computer can be connected to the same wifi hotspot (2.4g hotspot) at the same time.

③ Can use the serial port or BLE configuration module, and the module 2.4g must be connected to the antenna

#### 2、Start NetAssist.exe and open a tcp server:

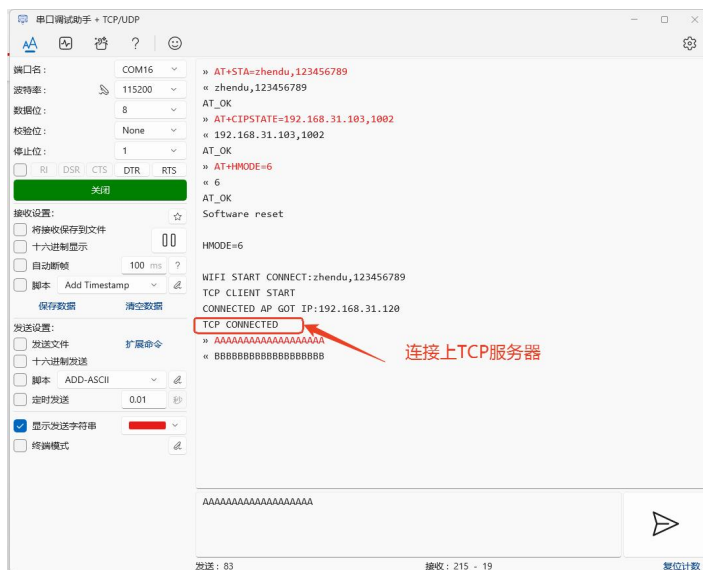


#### 3、AT command configuration to connect to TCP Server

AT+STA=zhendu,123456789 // set wifi account/password

AT+CIPSTATE=192.168.31.103,1002 // Set the IP/port of the TCP server to connect to.

AT+HMODE=6 // Enter mode 6, UART-TCP passthrough.

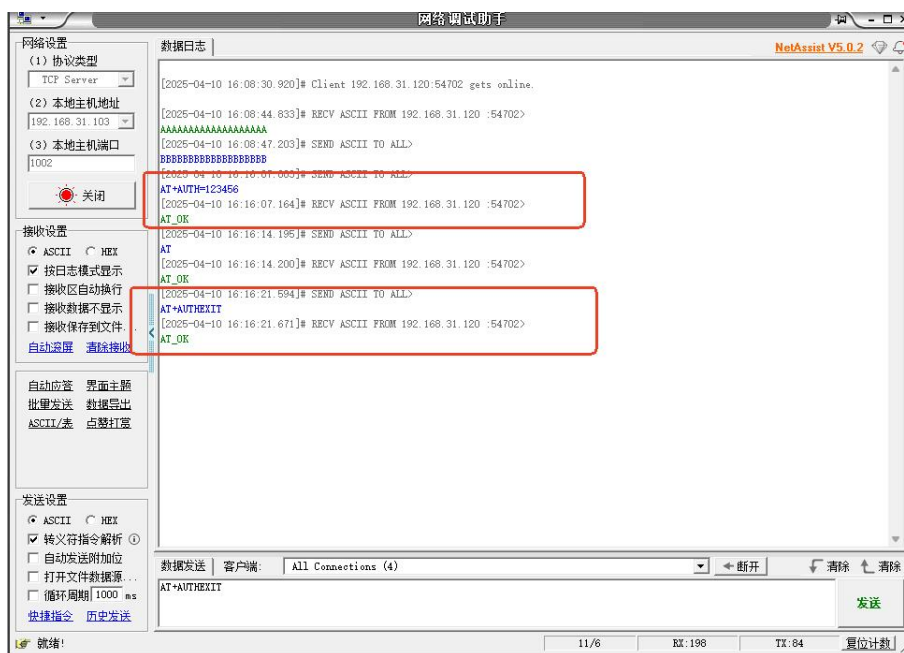


#### 4. Configured by NetAssist.exe using AT commands

AT+AUTH=123456 // Perform permission verification to enter AT command mode.

AT // Verify entry into AT command mode

AT+AUTHEXIT // exit AT command configuration



#### 4. LoRa fired at a fixed point

AT+HMODE=0 // Enter configuration

AT+TRANS=1 // Fixed-point transmission

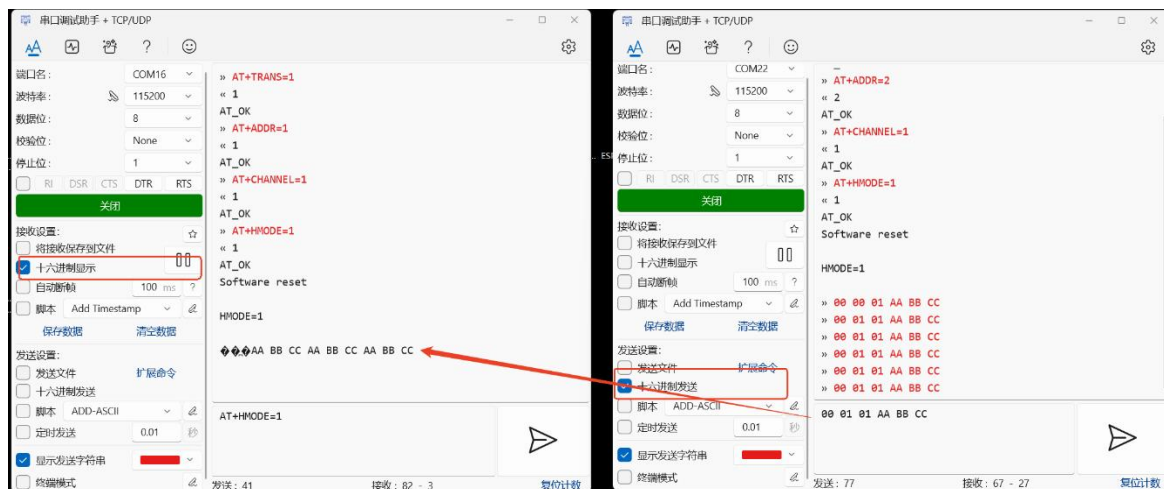
AT+ADDR=1 // Set module address to 00 01

AT+CHANNEL=1 // set channel to 1

AT+HMODE=1 // Set mode UART-LoRa pass-through, BLE listening

Transmit/receive data in hexadecimal

00 01 01 AA BB CC // Target address is 00 01, channel is 01, data is AA BB CC



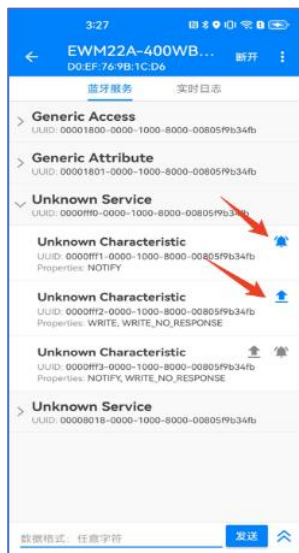
## 5. UART-LoRa Transmission, BLE Listening

### Topology:

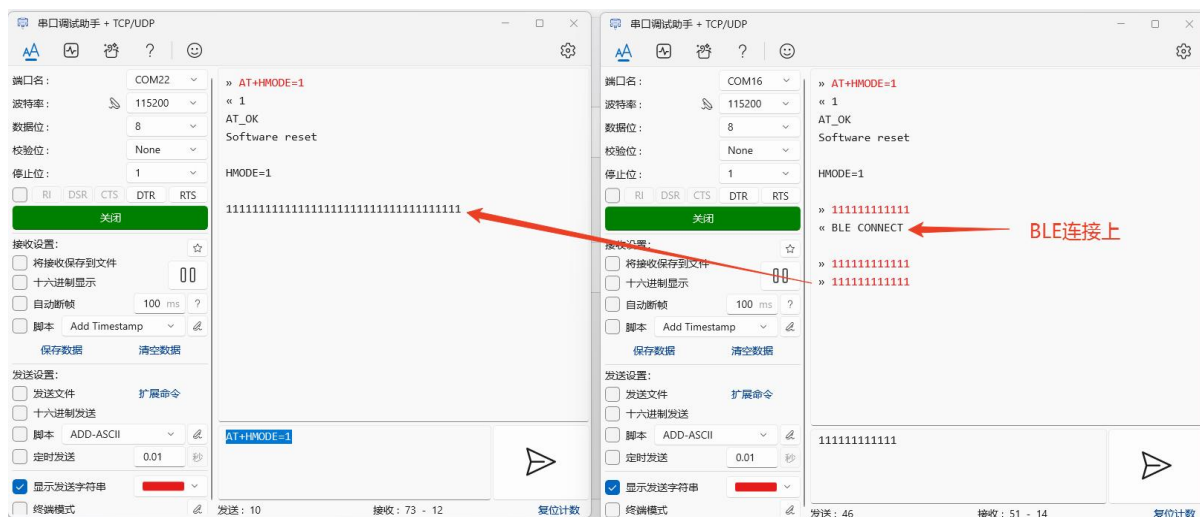
Two through LoRa to achieve serial port pass-through, the use of cell phones / or other Bluetooth host module (eg: E104-BT53) connected to one of the modules, to achieve the module to listen to the serial port and LoRa data, this example uses a cell phone to listen to the data

1, both modules serial port AT + HMODE = 1 command.

2、Use 【BLE debugging treasure】to connect to one of the module bluetooth, modify the MTU, through the FFF2 channel to send AT+AUTH=123456







## 6. UART-WIFI pass-through, LoRa listening

### Topology:

A module is connected to TCP to realize data passthrough between serial port and TCP, and the data flow will be sent out through LoRa, using another LoRa module can receive the data and print out through serial port.

### Module 1:

AT command to configure connection to TCP Server

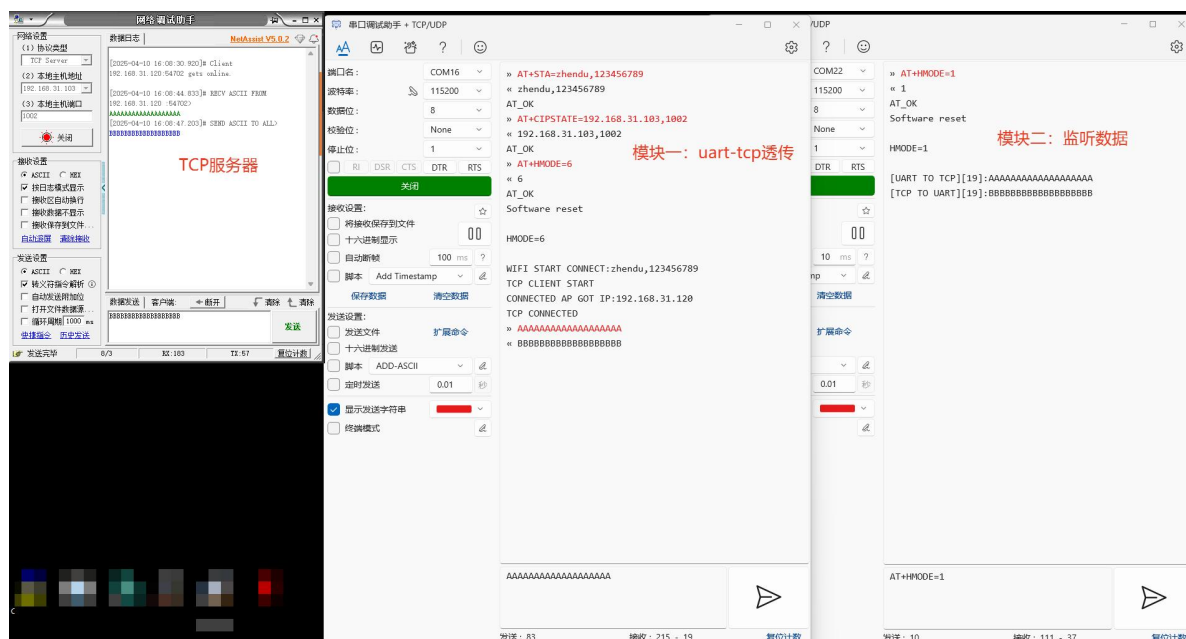
AT+STA=zhendu,123456789 // set wifi account/password

AT+CIPSTATE=192.168.31.103,1002 // Set the IP/port of the TCP server to connect to.

AT+HMODE=6 // Enter mode 6, UART-TCP passthrough.

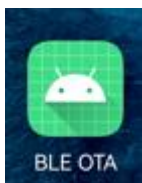
### Module 2:

AT+HMODE=1 // enter mode 1, UART-LoRa pass-through, BLE listening (mainly using LoRa)



## 7. BLE Upgrade module firmware

### 1. Preparation of cell phone APP (Android phone)



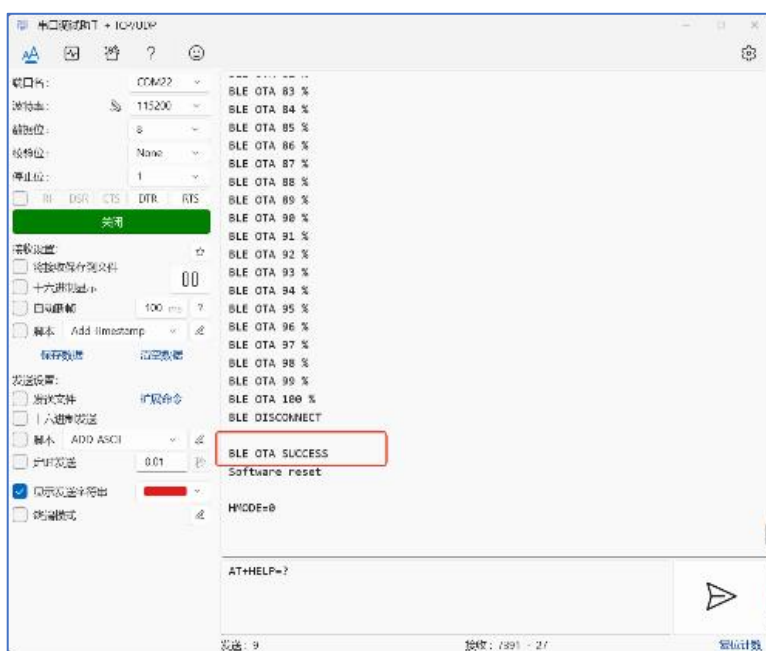
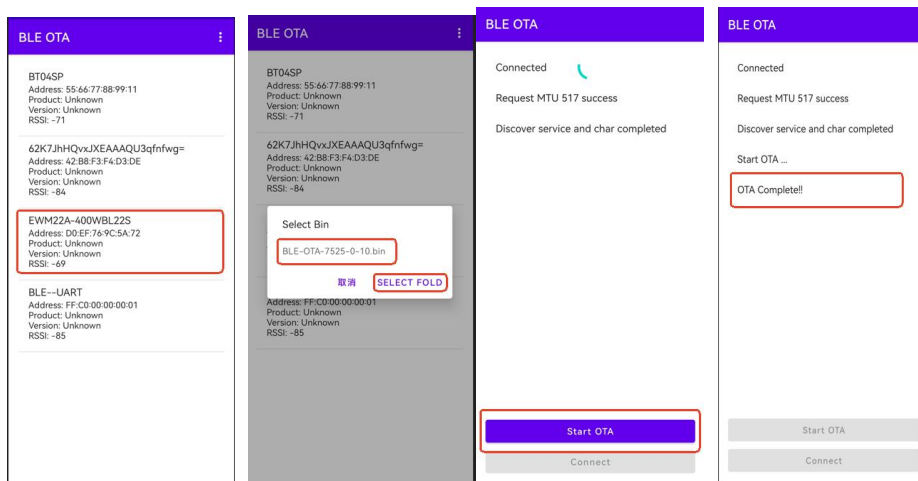
### 2. Prepare the upgrade file: BLE-OTA-7525-0-10.bin

### 3. Configure the AT command to enter BLE upgrade mode.

AT+BLEOTA=1 // Enter BLE upgrade mode.

### 4. Use the BLE OTA.apk upgrade program.

### 5. Connect the module's BLE, upgrade to select the firmware for upgrading.



## 7 Hardware design

- It is recommended to use a DC regulated power supply to power this module, the power supply ripple factor should be as small as possible, and the module should be reliably grounded;
- Please pay attention to the correct connection of the positive and negative terminals of the power supply, such as reverse connection may cause permanent damage to the module;
- Please check the power supply to ensure that it is between the recommended supply voltages, if it exceeds the maximum value it may cause permanent damage to the module;
- Please check the power supply stability, the voltage should not fluctuate significantly and frequently;
- When designing the power supply circuit for the module, it is often recommended to keep more than 30% of the residual amount, and there is the whole machine is conducive to long-term stable work;
- Module should be as far away as possible from the power supply, transformers, high-frequency alignments and other electromagnetic interference parts;
- High-frequency digital alignment, high-frequency analog alignment, power supply alignment must be avoided below the module, if you really need to go through the module below, assuming that the module is welded in the Top Layer, in the module contact part of the Top Layer paved copper (all paved copper and a good ground), it must be close to the digital part of the module and the alignment in the Bottom Layer;
- Assuming that the module is soldered or placed in the Top Layer, it is also wrong to route the module in the Bottom Layer or any other layer, as this will affect the spuriousness of the module as well as the reception sensitivity to varying degrees;
- Assuming that the module is surrounded by large electromagnetic interference devices will also greatly affect the performance of the module, according to the intensity of the interference is recommended to stay away from the module, if the situation permits you can do appropriate isolation and shielding;
- Assume that there is a large electromagnetic interference around the module alignment (high-frequency digital, high-frequency analog, power supply alignment) will also greatly affect the performance of the module, according to the intensity of the interference is recommended to be appropriate away from the module, if the situation permits you can do appropriate isolation and shielding;
- Communication line if you use 5V level, must be connected in series with 1k-5.1k resistor (not recommended, there is still a risk of damage);
- Try to stay away from some TTL protocols whose physical layer is also 2.4GHz, e.g. USB3.0;
- Antenna mounting structure has a big impact on the module performance, make sure the antenna is exposed and preferably vertically upward; when the module is mounted inside the chassis, use a good quality antenna extension cable to extend the antenna to the outside of the chassis;
- Antenna must not be installed inside the metal housing, which will result in a great weakening of the transmission distance.

## 8 Common problems

### 8.1 Unsatisfactory transmission distance

- When there are linear communication barriers, the communication distance will decay accordingly;
- Temperature, humidity, and co-channel interference, which will lead to higher communication packet loss rate;

- The ground absorbs and reflects radio waves, and the test results are poorer near the ground;
- Seawater has a strong ability to absorb radio waves, so the effect of the seaside test is poor;
- Metal objects near the antenna, or placed in a metal shell, the signal attenuation will be very serious;
- Wrong power register setting, air rate setting is too high (the higher the air rate, the closer the distance);
- Low voltage of power supply at room temperature is lower than the recommended value, the lower the voltage the lower the hair power;
- The use of antenna and module matching degree is poor or the antenna itself quality problems.

## 8.2 Modules are fragile

- Please check the power supply to ensure that it is between the recommended supply voltages, if it exceeds the maximum value it will cause permanent damage to the module;
- Please check the power supply stability, the voltage can not be substantial frequent fluctuations;
- Please ensure that the installation and use process anti-static operation, high-frequency device electrostatic sensitivity;
- Please ensure that the installation and use of the process of humidity should not be too high, part of the components for humidity-sensitive devices;
- If there is no special demand is not recommended to be used at too high or too low a temperature.

## 8.3 BER is too high

- Near the same frequency signal interference, away from the source of interference or modify the frequency and channel to avoid interference;
- Poor power supply may also cause garbled code, be sure to ensure the reliability of the power supply;
- Extension cords, feeder cords of poor quality or too long, can also cause high BER.

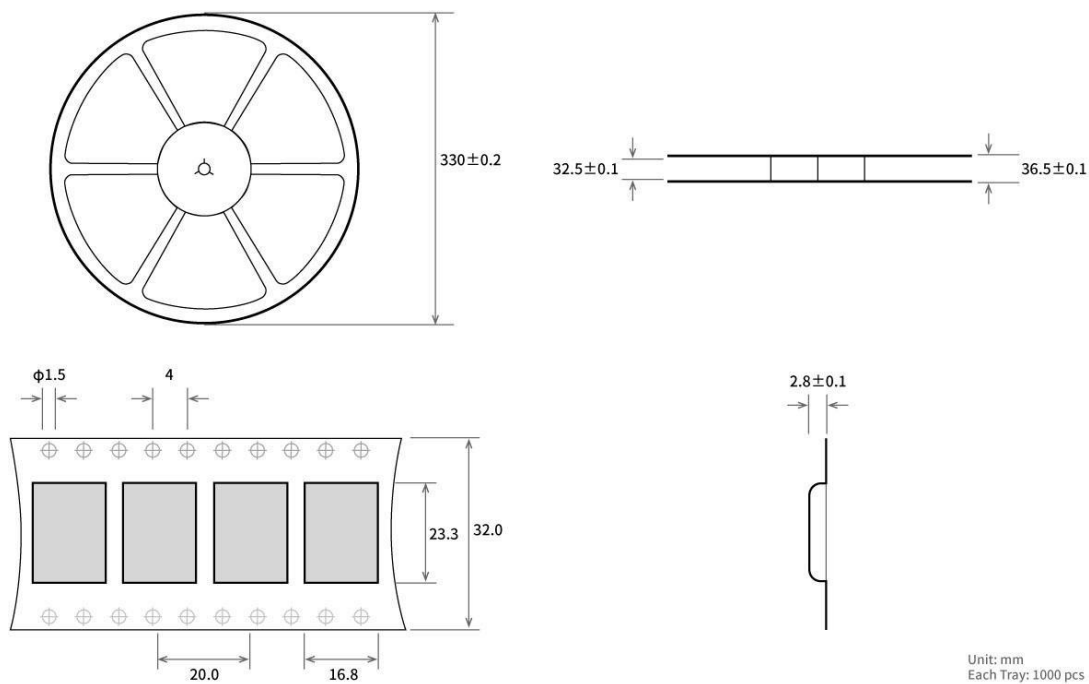
# 9 Welding instructions

## 9.1 Reflow temperature

| Profile Feature                                            | Curve Characteristics       | Sn-Pb Assembly | Pb-Free Assembly |
|------------------------------------------------------------|-----------------------------|----------------|------------------|
| Solder Paste                                               | Solder Paste                | Sn63/Pb37      | Sn96.5/Ag3/Cu0.5 |
| Preheat Temperature min (T <sub>smin</sub> )               | Minimum Preheat Temperature | 100°C          | 150°C            |
| Preheat temperature max (T <sub>smax</sub> )               | Maximum Preheat Temperature | 150°C          | 200°C            |
| Preheat Time (T <sub>smin</sub> to T <sub>smax</sub> )(ts) | Preheat Time                | 60-120 sec     | 60-120 sec       |
| Average ramp-up rate(T <sub>smax</sub> to T <sub>p</sub> ) | Average Rise Rate           | 3°C/second max | 3°C/second max   |



## 10 Product Packaging



## Revision history

| Version | Date      | Description      | Issued by |
|---------|-----------|------------------|-----------|
| 1.0     | 2025-5-13 | Initial version  | Bin       |
| 1.1     | 2026-5-15 | Content revision | Bin       |

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